Measured-State Flight Control for a Game-Physics Fighter:

INDI, Nonlinear Control Allocation and Damage Tolerance on the F-22E Strike Raptor and KR-67 Ifrit in *Nuclear Option*

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Development log, 21 September – 20 October 2026
F-22E FCS 1.4.0 – 2.2.23 · Ifrit FCS 0.1.0 – 0.2.1

Abstract

We describe a flight control system (FCS) written as a BepInEx/Harmony plug-in that replaces the stock fly-by-wire of Aryx's F-22E Strike Raptor mod for the combat flight game *Nuclear Option*, and its port to the game's stock KR-67 Ifrit. The game models each aerodynamic part as an independent plate with an airfoil lookup table, with no Mach effects on the airfoils, no tail blanking and no flap camber. The FCS is built on incremental nonlinear dynamic inversion (INDI) of the body rates. A Gauss–Newton control allocator works directly on per-surface nonlinear moment models evaluated at each surface's local angle of attack, so stalled and reversed surfaces are used in the direction that works. Around this core sit a G/pitch-rate command law with an AoA envelope, kinematic capture braking and protection; a velocity-vector roll law anchored to the yaw rate the airframe can deliver; and a layer of measured-state limits for asymmetric stores, fin loss and engine failure, ending in a VMC throttle limiter. A 6-DOF bench reconstructed from the mod's asset bundle and the game's own aero equations runs the unmodified FCS code. Over roughly 150 releases the worst pitch dip in the pull–release–roll manoeuvre fell from -65 to $-3.8\,^\circ/\mathrm{s}$, departures with one wing missing fell from 29/160 to 0/160, and the boundary of controllable flight with both fins lost was mapped (directional instability N_β from -0.1 to $-1.4\,^\circ/\mathrm{s}^2/^\circ$ over 10–50° AoA, best held-roll band 35–45°). We report what the allocator taught us about its own failure modes: cost terms that do not scale with dynamic pressure, stall plateaus, model mismatch that INDI cannot see, and saturation hidden by clamps. We also say where the remaining limits belong to the airframe and not to the controller.

1 Introduction

Nuclear Option (NO) simulates aircraft as assemblies of rigid parts joined by breakable joints. Each part carries its own aerodynamic plate. Aryx's F-22E Strike Raptor mod (v1.0.4) provides an F-22-like airframe:

- eight control surfaces: two all-moving stabilators, inner flaperons, outer flaperons ("ailerons") and two rudders on canted fins;
- two turbofans with pitch thrust vectoring (TVC);
- a stock FBW law.

That stock FBW is typical of the game:

- no real AoA or G command;
- no AoA protection;
- roll and yaw handled as separate axes;
- a roll law whose speed schedule never acts. Decompiled, its target is $10\,\mathrm{rad/s} \times \mathrm{clamp}(r_q, 0.5, 1)$ with a ratio r_q that always sits on the 0.5 floor: a fixed $286\,^\circ/\mathrm{s}$.

The goal was an FCS that flies like a modern relaxed-stability fighter, within NO's physics. The requirements:

- **Modes.** Gear down: rate command. Gear up: G and pitch-rate command. Stability Assist off: MPO.
- **Envelope.** AoA $+65/-50^\circ$ and $+9.5/-3.5\,\mathrm{g}$, with a pitch-rate budget of $48\,^\circ/\mathrm{s}$.
- **Pitch.** Released stick holds zero pitch rate. Easing the stick never shoves the nose.
- **Lateral.** Coordinated velocity-vector roll that becomes a single axis past 36° AoA, and 30–36 °/s of yaw rate in the falling leaf.
- **Effectors.** Pitch-only TVC, used to unload the stabilators so they stay free for roll and yaw.
- **Damage.** Graceful degradation when the jet is damaged.

A standing design preference was to *compensate from measured state rather than gain-schedule*.

This paper consolidates about 150 releases of notes, bench studies and in-game telemetry analyses into one account. The contributions are:

1. an INDI + nonlinear allocation architecture that works on a plate-based game aero model, including post-stall and reversed flow (§3);
2. pitch and lateral laws built from measured authority, not schedules (§4, §5);
3. airframe changes and what they did to static margin and roll (§6);
4. a damage-tolerance stack and the measured limits of a finless, engine-out F-22E (§7);
5. a catalogue of allocator failure modes found in flight, with root causes (§8);
6. the verification method (§9) and the Ifrit port (§10).

Units and evidence. Speeds are given in knots IAS unless stated otherwise. Bench numbers come from the 6-DOF reconstruction; in-game numbers come from telemetry CSVs. Bench numbers are reliable as relative changes and indicative as absolutes. Nothing in §7–10 after 2.2.15 has been flown at the time of writing unless a log is cited.

2 The simulation environment

The behaviour of the game's physics drove most of the design.

Per-part plate aerodynamics. Each `AeroPart` has a wing area, a drag area, an airfoil index, a centre of lift and a lift normal. Each step the aero job computes the part's local AoA from its velocity. It reads C_L and C_D from a 128-point airfoil table and applies the force at the part's rigid-body centre of mass, with the torque taken from the centre of lift. Aryx's mod uses three airfoils (Wing, LERX, Vstab). Every horizontal control surface uses the Wing foil, which peaks at $\pm 35^\circ$ local AoA. Three consequences follow:

- **No Mach effects on lift.** The static margin is identical at 194 and 486 kt.
- **No blanking.** A fin keeps working at any body AoA. Its local AoA is set by sideslip: about 0.6° at 45° body AoA.
- **No camber.** A deflected rudder or flap changes only its own plate's lift. On a real section the fixed surface ahead carries part of the extra force.

Thrust vectoring. NO's `Turbofan` zeroes the nozzle angles above 450 m/s *true* speed (875 kt TAS). At 16 km that is only about 325–350 kt IAS. Below the cut-off, nozzle pitch authority at 270 kt in full afterburner (AB) is about $\pm 52\,^\circ/\mathrm{s}^2$. The stabilators give about ± 150–$200\,^\circ/\mathrm{s}^2$ there.

Mass properties. Unity derives each part's centre of mass and inertia from its convex colliders at uniform density. The F-22E comes out at 29 230 kg (buffed loadout) with $I_{xx}, I_{yy}, I_{zz} \approx 111\mathrm{k}, 238\mathrm{k}, 340\mathrm{k}\,\mathrm{kg\,m}^2$. Inertial coupling into pitch is $(I_{zz} - I_{xx})/I_{yy} \cdot pr = 0.96\,pr$, which is large.

Airbrake. Aryx's speedbrake is eight `Airbrake` drag points (wing mid, wing rear, fins and stabs, each side). Each applies $0.6\rho v^2 \cdot$opening at its part's centre of mass: $9.6\,\mathrm{m}^2$ of drag area at full open. It also rotates the outer flaperons' lift normal by 15°. The inner flaperon and rudder deflections it shows are visual only.

3 Architecture

The plug-in runs beside Aryx's unmodified DLL. Harmony patches route the pilot inputs to the FCS, write the surface and nozzle commands into the game's control-surface jobs, and change the airframe (§6). Fig. 1 shows the loop.

3.1 INDI rate loops

The outer laws produce a body-rate command ω_{cmd} and from it a desired angular acceleration ν. INDI asks the effectors for the moment

$$M_{\mathrm{req}} = M_{\mathrm{ctrl}}(\delta_{\mathrm{now}}) + I\,(\nu - \dot{\omega}_f), \tag{1}$$

where $\dot{\omega}_f$ is the measured angular acceleration, filtered with a 0.04 s time constant. The unknown airframe moment cancels: whatever the plates, joints and gyroscopic terms are doing appears in $\dot{\omega}_f$. That is why an explicit inertial-coupling feed-forward turned out to be harmful. Evaluated at the *commanded* rates, it asked for 40–$90\,^\circ/\mathrm{s}^2$ of nose-down pitch the instant a roll began, for a coupling that did not exist yet. Turning it off (1.8.14) took the mean worst pitch dip in pull–release–roll from -25 to $-13\,^\circ/\mathrm{s}$. INDI measures the real coupling as it develops.

Hybrid INDI (1.9.0). At 700 kt the airframe's own pitch moment changes quickly with AoA and q, and the filter lag let q overshoot its command by about $2\,^\circ/\mathrm{s}$ for 0.3 s, which is about 1.4 g at that speed. The lag is now predicted from model slopes of the non-control parts, re-evaluated every 0.1 s:

$$\Delta M_{\mathrm{lag}} = \tfrac{\partial M}{\partial \alpha}(\alpha - \alpha_f) + \tfrac{\partial M}{\partial q}(q - q_f). \tag{2}$$

The term fades out between 25 and 40° AoA, where it cost roll onset. G peaks for a 3 g command at 700 kt fell from 4.49 to 3.25 g.

3.2 Nonlinear control allocation

Each effector i has a model $M_i(\delta_i, x)$ of its three-axis moment. It uses the game's own plate equation and airfoil table at that surface's local AoA, sideslip and dynamic pressure, with the current deflection limits. The allocator minimises

$$J(\delta) = \left\| W\left(\textstyle\sum_i M_i(\delta_i) - M_{\mathrm{req}}\right)\right\|^2 + s(\bar{q})\left[\mu \sum_i c_i \delta_i^2 + \lambda \sum_i (\delta_i - \delta_i^-)^2 + J_{\mathrm{pair}}\right] + J_{\mathrm{fight}} \tag{3}$$

by Gauss–Newton with box constraints (travel and per-frame servo reach). The Jacobians are numerical, taken on the grids. W weights the axes. The first bracket holds the neutral preference (μ) and the move cost (λ). J_{pair} prices the symmetric and antisymmetric components of each pair, $\frac{1}{2}(\sigma_i\delta_i \pm \sigma_j\delta_j)$ with pair signs σ, so differential stab and rudder toe-in can be priced on their own. TVC is the cheapest pitch effector: its move and position costs are about $50\times$ lower than symmetric stab. That is how the design intent "TVC unloads the stabs" is expressed.

Stall- and reversal-aware models. Because the M_i are evaluated on the real lift curve, a surface past its peak is deflected the way that still produces moment. Two exceptions were needed:

- **Pitch.** The stab pitch model is held monotone in the normal direction (1.4.0). Using a stalled stab "backwards" for pitch left it parked at $+20$–28° after high-AoA releases.
- **Reversed flow.** When a stab's local AoA passes 120° (hysteresis to 110°), the monotone model follows the reversed sense (1.7.6). In a tail slide the air arrives over the trailing edge. The monotone model then read zero pitch authority, and the jet shoved nose-down at -53 to $-84\,^\circ/\mathrm{s}$ while the law commanded zero. After the fix, nose-down time in all 18 held tail-slide cases was 0.0 s.

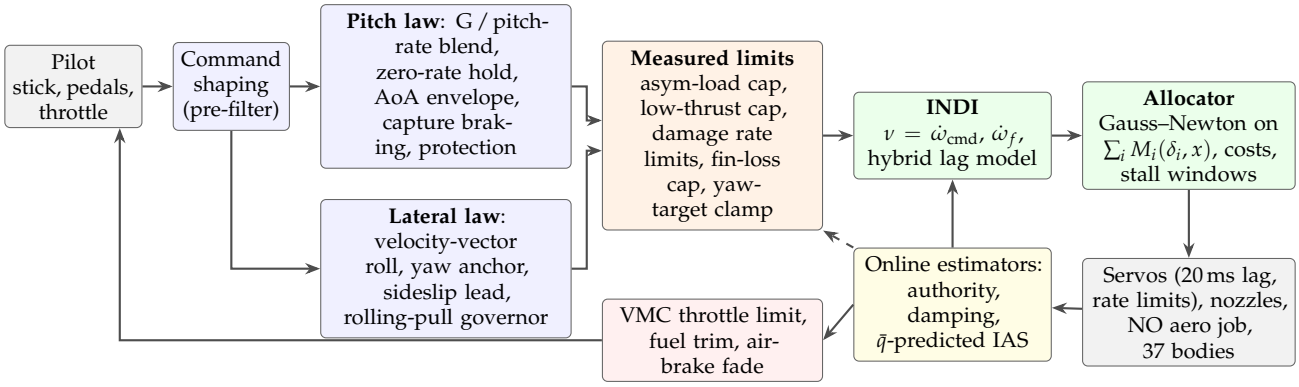


Figure 1: FCS structure. Every limit and authority figure is computed from the live airframe state: surface models at local AoA, live nozzle thrust, and measured accelerations. Nothing comes from an offline schedule.

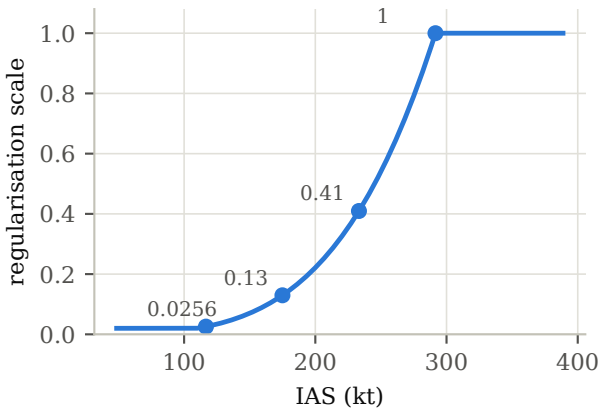


Figure 2: Allocator regularisation scale versus IAS (1.8.0). Reference $\bar{q}$ at 292 kt; floor 0.02.

Low-$\bar{q}$ relief (1.8.0). The error term is a squared moment, so it scales with $\bar{q}^2 \propto V^4$. The regularisation is in travel units and does not scale. At 117 kt the error term carried $(60/150)^4 = 2.6\%$ of its weight at the 292 kt tuning point. The solver's cheapest answer was to stay near neutral, and that showed up as weak roll and slow reversals. The fix scales the regularisation by $s(\bar{q}) = \text{clip}((\bar{q}/\bar{q}_{\text{ref}})^2, 0.02, 1)$ (Fig. 2). High-speed tuning is unchanged bit for bit. This was a units bug present since the allocator was first written.

Fight costs. J_{fight} penalises groups (stabs, outer flaperons, inner flaperons) whose roll opposes a reference sign. It was introduced for pedal inputs (1.8.41) and extended to released stick (2.2.16) to break crossed-set locks (§8). The reference is the groups' net roll, or the sign of the roll target when $|\text{target}| > 15\,^\circ/\text{s}^2$ (2.2.17). With the stick released, it waits until the pitch has settled. For the stabs it counts only the roll of their *split* about the common deflection (2.2.21).

Paired stab/TVC search and stab rate bound (1.8.14–1.8.19). The nozzles reach a new position in about 0.1 s; the stabs need about 0.3 s. If the allocator places the nozzles to balance where a stab *will* be, the nozzles push the wrong way for 0.2 s. With the stick released, each allocation may therefore command the stabs only as far as their servos can travel in 0.1 s. A paired search hands pitch from a stab parked past its useful range to the nozzles, so the stab goes back to roll and yaw. With pitch and lateral stick held together, the stabs are shared out in proportion to lateral stick (`LateralStabReserve` 0.7). The nozzles carry what pitch they can, and with no lateral stick pitch keeps full authority.

4 Longitudinal control

4.1 Command laws

The pitch-rate budget follows the SimplePlanes design that the pitch logic was ported from. It is IAS-based at low AoA and T/W-based otherwise, with a maximum of 48 °/s. Three laws were shipped:

- **1.4–2.0: AoA/G command,** blended by β_G, the reachability of the G limit within the rate budget.
- **1.9.0: AoA-to-rate hand-over.** Past an AoA threshold the stick becomes a pitch-rate command. The threshold is 36° while the wing alone can lift 2× the weight, $\bar{q}SC_{L,\max}/mg \geq 2$, and falls to 25° as that ratio falls to 1. The capture brakes the jet onto the threshold at the rate it will carry through it, so q is continuous across the hand-over.
- **2.1.0, the default: G + pitch-rate command.** Pilot feedback was that AoA command is poor for gun tracking even at low AoA. An AoA-then-rate-hold hybrid was tried in 2.1.8 and rejected. The stick commands pitch rate where full G is out of reach and G where it is in reach. AoA survives only as a *limit*.

Released stick and easing. Released stick holds zero pitch rate. Easing the stick targets the pitch rate the aircraft will sustain at the new AoA: the present flight-path rate plus the lift change taken from the envelope sweep. The jet reaches it through a jerk-limited deceleration, so the nose never reverses. With the stick released, a pitch priority multiplies the pitch weight in W by up to 30 once q leaves the hold (1.8.14). Without it, the allocator would trade the arrest of a 40 °/s pitch rate for roll moment on the stab pair.

Command shaping (1.9.0). Small AoA and G steps are built gently and large ones fast. The pre-filter bandwidth runs from $1.5\,\mathrm{s}^{-1}$ for a tiny step to $10\,\mathrm{s}^{-1}$ at 15° (or 3 g) and above, and the step size is held so a large step keeps its speed all the way in. The idea follows the self-adjusting pre-filter gain reported for the T-50 control law. AoA scatter in a noisy 26° hold fell from 0.096 to 0.021° standard deviation.

4.2 Envelope, capture and protection

Kinematic capture. The approach to a limit is braked using the nose-down authority the jet really has at that AoA, with the nozzles at live thrust (1.8.20). Mean / worst capture overshoot went from 0.8 / 3.8° to 0.0 / 1.5° in AB, and from 1.7 / 5.5° to 0.0 / 1.6° at military power. An integrator anti-windup brought the steady-hold bias to −0.06...0.00°.

Low-thrust cap (1.9.0). Every 0.2 s the FCS sweeps the nose-down authority available from full nose-down stab, flaperons at trim and nozzles at present thrust. The AoA limit is set 1° below the AoA where that authority runs out, with an $8\,^\circ/\mathrm{s}^2$ margin. Bench no-return AoA at zero thrust is 24–28°. A full pull then full push at idle used to reach 80° and not recover; with the cap it peaks at 24.6° and recovers. The same sweep explained a pitch-up seen in game at high altitude. Above the TVC cut-off the stabs alone cannot hold more than about 20–25°, and the AoA ran to 74–88° before the cap existed.

Protection. Past the limit, protection drives the nose down at the full $48\,^\circ/\mathrm{s}$ whatever T/W is, scaled by $(1 - \text{stick})$, so full aft stick cancels it. Two bugs were found in the override:

- In a tail slide AoA wraps from +179° to −179°. The wrap read as a −45° violation and shoved the nose at $-48\,^\circ/\mathrm{s}$. AoA is now unwrapped through ±180°.
- As AoA fell back through the limit, the hold handed back to the normal law, which then chased the falling flight path. The override is now latched until the stick is eased below 90%.

Negative limit. Inverted corkscrews logged in game at −65° and −60° departed at 68–97 kt with 78–83° of sideslip. The FCS commanded the right sideslip correction ($\dot\beta$ of −6 to $-17\,^\circ/\mathrm{s}$) and got +10 to +13. On the negative side the canted fins sit deep in post-stall, at fin local AoA −68 to −79° for body −60 to −65°. Fig. 3 shows the bench matrix. Only −50° held for both the light and the heavy (37.6 t, T/W 0.89) jet, so it became the default. This is an airframe limit, not an FCS bug.

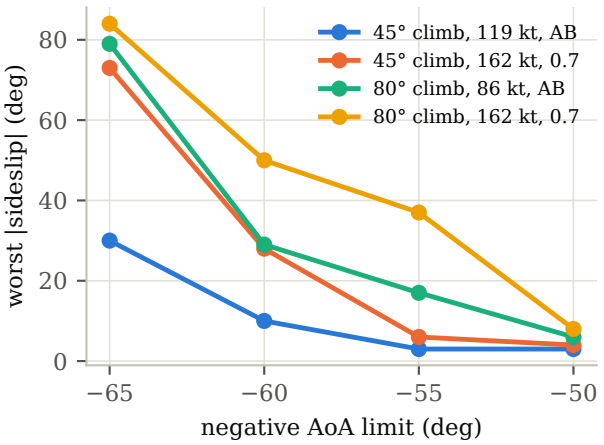


Figure 3: Worst sideslip in a 25 s full-push + full-lateral corkscrew versus the negative AoA limit (bench, light jet, entry speed and throttle as labelled). Climb = entry flight-path angle.

5 Lateral–directional control

5.1 Velocity-vector roll and its yaw cost

The lateral law commands a stability-axis roll rate p_S that blends into a single axis past 36° AoA. Coordinating it needs a body yaw rate $r = p_S \sin\alpha$. The sideslip kinematics,

$$\dot\beta = p\sin\alpha - r\cos\alpha + \tfrac{g}{V}\cos\theta\sin\phi, \qquad (4)$$

make the yaw axis the bottleneck at AoA. Every roll limit therefore comes from *measured yaw*:

- **Roll ceiling (1.4.0).** An online estimate of roll damping (every aero part, with a small roll rate added) and of usable roll control moment gives the full-deflection steady rate. `RollStopFactor`$\times a_{\mathrm{roll}}$ makes the ceiling follow inertia, for example with heavy stores (1.8.0).
- **Yaw anchor (2.1.0).** The roll target is clamped so the yaw rate it needs stays within $r \pm 0.5\,a_{\mathrm{yaw}} \cdot 0.25$ s of the present yaw rate. During a full push the AoA swept from +15 to −45° under a held roll, and the unbudgeted term $p_S \cos\alpha\,\dot\alpha$ drove sideslip to 31–50°. With the anchor, the worst sideslip is 3.2–6.1°.
- **Sideslip-rate lead.** The predicted sideslip is $\beta + 0.8\,\dot\beta_{\mathrm{kin}}$. The roll backs off from 4° of predicted sideslip and reaches zero at 14°, but only while the sideslip is still growing.
- **Rolling-pull governor (2.2.7).** A nose-up AoA-rise governor, active from $10\text{–}20\,^\circ/\mathrm{s}$ of velocity-vector roll, bounds the AoA rise by what the measured yaw authority can follow and by a sideslip margin (adjustable 2–15°). At its original 6° default, rolling pulls went from 14.3 / 20.5° to 4.1 / 5.3° mean / worst sideslip, at the price of about 3× slower AoA build. Since 2.2.23 the default is 9° with a roll trade of 0.7, which slows the roll so the AoA can rise inside the margin: full aft + full roll from wings level reaches 20° in 1.4–1.8 s (before: not within 4 s at 178–297 kt) with 5.2° worst sideslip, for a roll of about $44\,^\circ/\mathrm{s}$ instead of 149.

5.2 The rudders at high AoA

A static sweep of the allocator's surface models (Fig. 4) changed the high-AoA design:

- The canted rudders' inverse roll is present at every AoA.
- Through $\dot\beta$ that inverse roll *adds* to the rudders' yaw effect on sideslip.
- Past about 55° the rudders are almost a pure sideslip

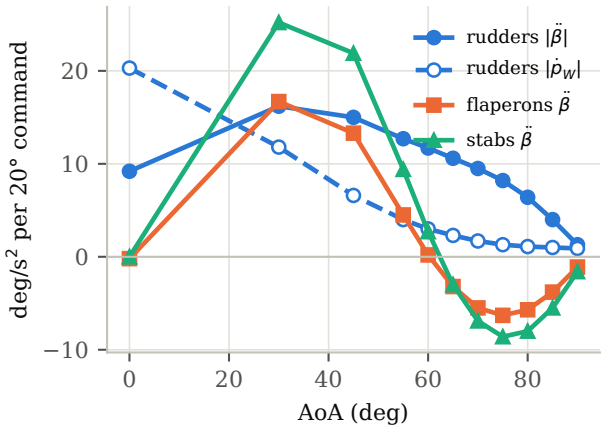


Figure 4: Sideslip ($\ddot{\beta}$) and velocity-vector roll ($\dot{p}_W$) authority per 20° lateral command versus AoA, 135 kt, $\beta = 0$ (allocator surface models). The ratios do not depend on speed.

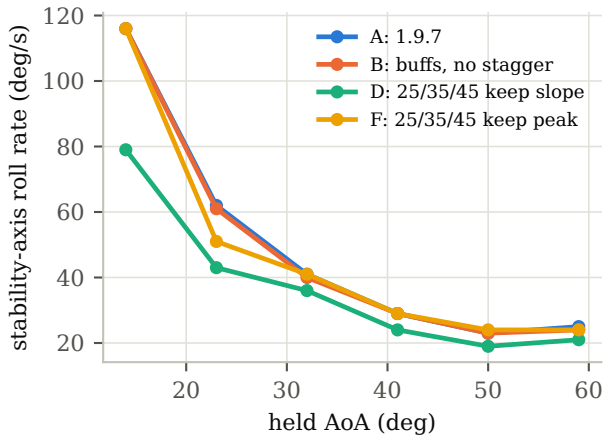


Figure 5: Stability-axis roll rate at held AoA, 189 kt, full lateral stick, AB, for four airframe configurations (bench).

effector: $|\ddot{\beta}|/|\dot{p}_W|$ is 3.2 at 55° and 6.1 at 75°.

- Between 55 and 70° they are the *only* good sideslip effector. The flaperons' and stabs' sideslip authority collapses at 60° and reverses past about 62°.

The original 35–50° rudder fade was justified as tail blanking, which NO does not model. All it did was take yaw away. With it gone, yaw came from differential stab, which makes 7 °/s² of roll per 1 °/s² of yaw; the rudders make 2.4. Moving the fade to 50–60° (2.2.0) cut the worst sideslip at the 45° limit from 13.4 to 5.4°. Reversals got up to 1.2 s quicker.

Fin camber coupling (2.2.0). NO flies a rudder as its own plate. In a real flapped section the deflection cambers the whole fin. Thin-airfoil flap theory for a half-chord rudder gives about 0.75 section effectiveness, and the plate already carries its share, which leaves $K \approx 0.5$ for the fin. The fin force at local AoA $+K\delta_r$, minus its force at local AoA, is added as a correction force at the fin's centre of lift, and the same term goes into the FCS models. Yaw authority rises about 25% and the opposite roll about 55%. At high AoA that roll also cuts sideslip.

Falling leaf and low-IAS feel. The requirement was 30–36 °/s of yaw in deep stall. The low-IAS yaw margin became a curve (1.7.0): full ×1.35 at or below 93 kt, decaying with a 19 kt scale. A flat margin up to 233 kt had only let the roll command outrun the jet's ability to stop it. Falling-leaf yaw rates reached 26–35 °/s on the bench.

6 Airframe modifications

All F-22Es, AI included, receive these changes. The thrust, drag, fuel and countermeasure buffs are a separate profile toggle; the stock profile is the default.

Static margin (1.5.0). Raising stab and flaperon effectiveness to 1.3 and 1.5 had moved the neutral point from 0.20 to 0.51 m aft of the CG. That made the jet *more* stable and increased trim drag. The re-balance:

- stab 1.1, LERX 1.6;
- CG moved 0.05 m aft by transferring 158 kg of part mass;
- static margin 0.09 m at 0–10° and 0.08 m at 10–30°;
- trim at 21° / 311 kt went from stab 9–10° + TVC 10° to stab 4.7° + TVC 1°.

LERX at 1.75 produced a nose-up break at 40–45°, and at 2.0 the whole airframe was unstable. The values were later revised to flaperon 1.2, LERX 1.2 and stab 1.2 (2.0.0–2.0.3).

Staggered stall (2.0.0). Every horizontal surface used the same 35°-peak foil. Each family now gets a remapped copy, injected into the aero job's global chart array: stabs peak at 25°, inner flaperons at 35° and outer flaperons at 45°. In the default keep-peak-C_L mode the stab lift slope is 1.4×. Fig. 5 shows the negative result: *staggering does not add roll authority*. The roll dip comes from a pair parked on its own lift peak, where one side deflects up the curve and the other down and the moments cancel. Moving the peak only moves the dip. Keep-slope mode lost 15–30% of roll everywhere. The stagger was kept because it improves low-AoA static stability (neutral point +0.29 m at 0–14°). Its price is a nose-up tendency past 30° with the stabs at neutral, which the FCS holds.

Other changes.

- Pitch-only TVC, with nozzle travel ±20° from 1.8.27.
- Servos at 100 °/s behind a 20 ms first-order lag, which also cut a rolling-pull overshoot at 778 kt from 10.8 to 9.8 g.
- Outer flaperons droop with the flap schedule.
- Slats held out to 70° AoA.
- Stab downwash correction (2.0.8).
- Ground aero-braking: trailing-edge-up direction found per surface from the model; rudders toed in with zero antisymmetric yaw by construction.
- Fuel transfer to trim lateral CG (2.0.12).
- An in-flight airbrake that fades with its share of total drag. The share is 81% at 0° AoA and 9% at 50°; the

brake is full above a share of 0.4 and stowed below 0.15 (2.2.16). That frees the flaperons and rudders at high AoA for 5–12% less deceleration.

7 Damage and failure tolerance

The FCS assumes nothing about which parts survive. Every 0.2 s an asymmetric-load sweep evaluates the live airframe. The response is layered.

1. Asymmetric loads. As the measured asymmetry grows, the AoA limit is capped and the stick is remapped onto the cap, keeping enough lateral authority for the pilot. The roll limits use the authority left *in each direction* after the damage trim (`AsymRollHeadroom`, 1.8.17). With one wing missing, bench departures fell from 29 of 160 cases to 0 of 160 and standing sideslip from 2.5 to 0.3°.

2. Lost yaw authority (2.0.12). With a fin gone, roll and yaw rate limits scale with the share of rudders left (never below 0.35). The yaw target is clamped to what the remaining effectors can reach. Before this, after a fin loss the allocator chased −50 to −106 °/s² of yaw with about −12 available. It thrashed one stab between +35 and −30° at 0.6 Hz, and pitch rate swung ±40 °/s. Afterwards, stab rate rms fell from 53–61 to 5–6 °/s.

3. Decelerons (2.2.7, 2.2.19). With both fins gone the clamp was zero, so no yaw was requested at all. Past the rudders' reach, the inner and outer flaperons on one wing now open against each other: their rolls cancel and their drag yaws the jet. Across 16 bench cases with both fins detached, sideslip went from 19.8 / 88.1° to 8.7 / 16.7° mean / worst. All four flaperons were already at ±25° in held rolls, so the split may use their full drag yaw (`SplitFlaperonYawUse` 1.0).

4. Fin-loss directional stability. Regressing the unexplained yaw acceleration on $\beta(V/V_0)^2$ in game logs gave the finless F-22E's N_β (Fig. 6): directionally unstable from about 10°, falling to −1.4 °/s²/° near 50°. The bench agrees with r^2 0.9–1.0. The bench matrix (Fig. 7) shows a non-obvious structure:

- below 35° a held roll departs, because the velocity-vector roll needs yaw rate the decelerons cannot make;
- 35–45° is the best band;
- above 50° nothing departs, but sideslip grows.

`FinLossAoACap` 40 phases in from one fin lost to two. It brought held-roll sideslip from 13.4 / 21.9° to 4.6 / 10.4°.

5. Engine out and VMC (2.0.12, 2.2.22). An engine counts as out when it is inoperable, or when it makes under 15% of the other's thrust while that one makes more than 20 kN. For the engine-out AoA cap, only the aero part of the yaw load is scaled with the dynamic pressure the pull is heading for; the thrust part does not fall with speed. The cap then tightens at 40 °/s. The VMC limiter is the last resort. When an engine is out, yaw is saturated and sideslip is running away, it pulls the throttle back at 0.8/s. With the fins intact, the rudders' yaw authority equals the AB thrust asymmetry at about 124 kt.

With both fins gone the limiter never acted. Saturation was judged against the clamped yaw target, so the target always read as reached. Counting the clamped-away demand as saturation fixed it (Fig. 8). Engine out with no fins is now controllable from 324 kt up; at 216 kt and below it still departs from part throttle. The flaperon drag yaw grows with $\bar{q}$ but the live engine's thrust yaw does not, so the remaining limit is the airframe.

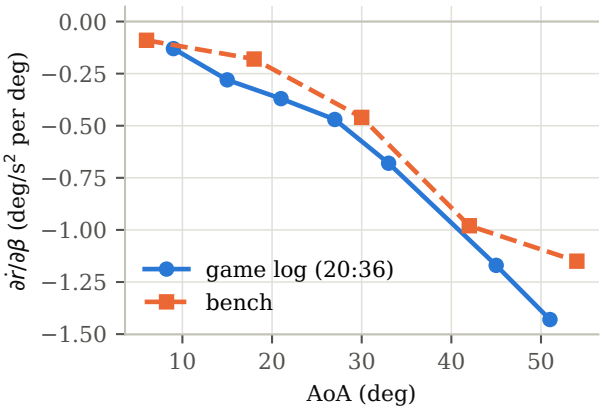


Figure 6: Weathercock derivative of the F-22E with both fins lost, referenced to 161 kt: game telemetry versus bench.

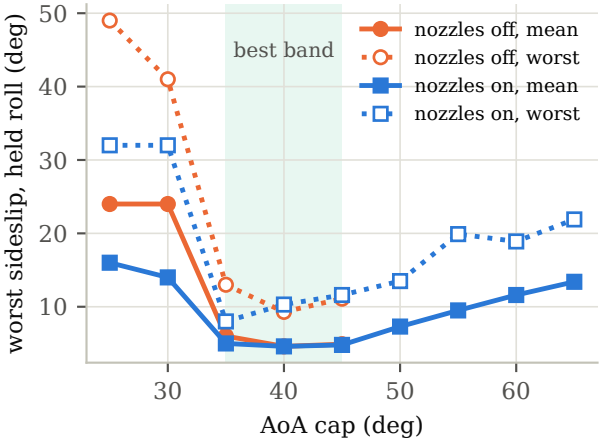


Figure 7: Worst sideslip in a held full roll versus the AoA cap, both fins gone (bench, 81–216 kt, throttle 0.5/1.0). Without TVC the bench jet cannot exceed about 40°.

8 Allocator failure modes

Most of the hard bugs were not in the control laws. They were in how the allocator's cost function interacted with the plant. Each item below was found in flight telemetry, reproduced on the bench and fixed at the cause.

F1. Unscaled regularisation (§3, 1.8.0). Costs that do not share units with the error term change meaning with $\bar{q}$.

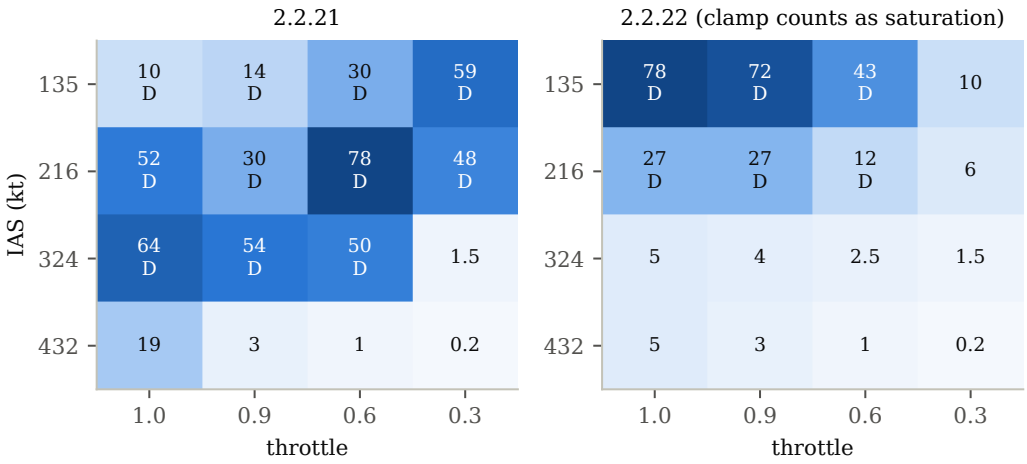


Figure 8: Both fins detached, left engine fails 0.5 s into the run, stick centred 10 s: mean sideslip over the last 3 s (deg). D = departed. Left: the VMC limiter blind behind the yaw-target clamp. Right: clamp excess counted as saturation.

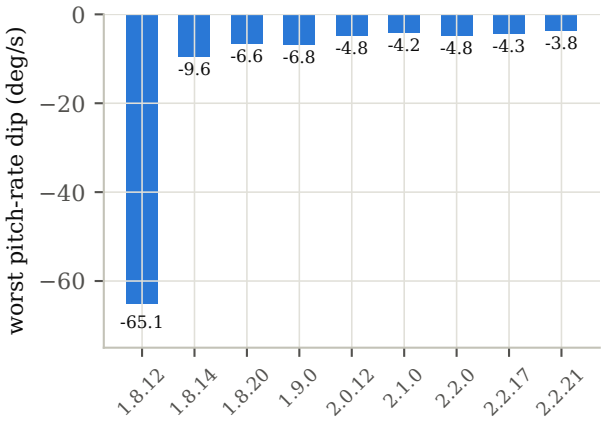


Figure 9: Worst pitch-rate dip in the pull–release–roll set across versions (bench). 1.8.12–1.8.14 used the 64-case set at 243–324 kt; later versions use suite S1. The sets are similar but not identical.

F2. Trading pitch for roll after a release (1.8.14). The stick was released while the jet was arresting a 40 °/s pitch rate, and the solver accepted a pitch miss to get roll moment. One stab went to its −30° stop and was then 20° and more than 0.2 s away from where pitch needed it. With no outer loop to recapture, the error became a −37 °/s nose shove. Fix: released-stick pitch priority and the stab rate bound. Fig. 9 tracks this manoeuvre across versions.

F3. Chatter under saturation (2.0.12). When the yaw target is out of reach, the solver jumps between two near-equal solutions faster than the servos can travel. Fix: clamp the target to what can be reached.

F4. Stall plateaus (2.2.15, 2.2.18). On its post-stall plateau a stab's modelled moment barely changes with deflection, so moving it back shows no gain. A left stab stayed at −30° at 31–35° local AoA for 15 s with the stick centred. Everything else sat against it at the stops and the roll-trim integrator wound to −189. With TVC on, the nozzles took up the slack, which is why no TVC-on log showed it. Fix: with the stick centred and below 30° AoA, `StabParkEscape` confines each stab to the attached side of its curve. In 2.2.18 a second bug was found: the window had searched only within one frame's servo reach, so a stab parked further out was never walked back.

F5. Crossed sets from model error (2.2.16). INDI cancels *airframe* moment error, but it cannot see an error in the *control-effectiveness* model. The logs showed the game's stab split producing only 60–80% of the modelled roll, a regression error of +3.1 against a model derivative of −7 to −8.4 per degree·$(V/194\,\text{kt})^2$. With the stick centred, the pull to neutral moves the surfaces along the *model's* zero-moment direction, which leaks real roll. The allocator then corrects with its cheapest roll effector, the overrated stab split. Each cycle crosses the set further, until the wing and rudders sit at their stops against the stabs. The sum's modelled roll was +200 °/s^2 against a measured 0. A bench plant/model mismatch knob (plant stab lift ×0.8) reproduced all three in-game locks. The fight cost with released stick fixed them. The source of the mismatch is still unidentified.

F6. A cost term's side effects (2.2.16–2.2.21). The fight cost introduced three problems of its own:

- **It fought roll stops.** At release the net roll still has the old sign. Fix: take the reference sign from the roll target.
- **It priced the nose-down split at release.** Fix: wait for the pitch to settle.
- **It froze the stabs in sideslip with the fins gone.** With sideslip present the two stabs see different flow, so even a pure *common* (pitch) stab motion rolls the jet, and it was priced as a fight. The stabs froze for up to 328 frames while the nozzles walked to ±20° and the zero-rate hold drifted. This was first misattributed to TVC; the replay showed otherwise. Fix: count only the roll of the stab split. Pitch-rate

error rms fell from 1.27 to 0.46 °/s, matching the fins-intact replay (0.50).

F7. Saturation hidden by clamps (2.2.22). Any consumer that judges saturation against a target already clamped to what is reachable will never see saturation. The VMC case above is an example.

F8. Pitch side effects of roll. Rolling at low AoA with TVC on, the nozzles swung from $+4$ to $-4°$ while q stayed within ± 0.4 °/s. The cause was the 71 kNm of pitch moment produced by roll deflections on nonlinear parts of their curves, which the allocator cancelled with its cheapest effector. That is correct trim work, not a fault. `TvcNeutralHold` (2.2.14) moves the steady part onto the stabs when the pitch rate is held.

Lesson. Every cost term in a nonlinear allocator needs three things: consistent units with the error term, an explicit statement of which state it must *not* act in, and a test with a deliberately wrong model. Several of these failures were bistable. One release case landed on either branch depending on a 3 kt change in entry speed. Release sets are now flown at three speeds about 3 kt apart (96 cases) so that a version comparison does not pick up a branch flip.

9 Verification method

Bench. From 1.8.14 the bench has flown the real jet. Its parts:

- **Airframe.** The transform hierarchy and all 37 `AeroParts`, extracted from Aryx's asset bundle, with mass properties from the collider hulls.
- **Aero and actuation.** The three airfoils; the eight surfaces with ranges and servo speeds; the engines, nozzles, high-lift devices and flap drives.
- **Physics.** NO's own `AeroJob_Math`, `ControlSurfaceJob_Math` and nozzle slew, re-implemented.
- **FCS.** The shipping `FcsController` and allocator, compiled unmodified against managed Unity stubs. The stubs are generated from the plug-in source and the pilot's live config.

The bench reproduced the logged weight exactly (29 230 kg, "235 kg moved Fore → Back"), the logged departures, and the finless N_β curve.

Test sets.

- Suite S1–S7: pull–release–roll, held-pull rolls, 1 g rolls, post-stall and falling leaf, reversals.
- Hitch, release, rolling-pull and roll-onset sets.
- AoA-cap and engine-out matrices.
- *Replays* of logged stick inputs, the most direct test of a reported fault.
- Plant/model mismatch knobs, part detachment, store and mass overrides.

Telemetry. The FCS writes a wide CSV: commands, estimates, authorities, every surface and nozzle, and debug columns per feature. Regressions on it produced the N_β curve and the stab-split mismatch. Kinematic sideslip-rate reconstruction matched measured $\dot{\beta}$ to about 1 °/s.

Limits. The bench is a single rigid body, while the game uses 37 jointed bodies. It assumes full fuel unless overridden. Some in-game behaviour does not reproduce: the stab-split mismatch, and the second roll in the 20:11 log. Where bench and pilot disagree, the pilot has been right more often than the bench.

10 Port to the KR-67 Ifrit

The Ifrit is the game's own multirole jet: two all-moving canted tails (about 40° dihedral), fins with no rudders, ailerons, split flaps and 2D nozzles with pitch *and* roll vectoring. The port (0.1.0) kept the 2.2.5 control laws unchanged and added two effector types:

- **split decelerons as yaw effectors,** where the yaw comes from drag $F = 0.1 \cdot \text{split} \cdot \rho v^2$;
- **differential TVC** about the symmetric nozzle angle.

Bench findings:

- The airframe is nose-down stable at every positive AoA, so pitch recovery is never short; the pitch law ported unchanged.
- Coordinated roll at high AoA is starved of yaw. At 135 kt and 45° AoA the velocity-vector roll ceiling is 15 °/s² from surfaces alone, 22 with the split and 31 with roll TVC.
- A loaded roll at the 65° limit gives 10 °/s with 7° of sideslip on the F-22E's effector set, and 30 °/s with 1° once the split and roll TVC are added. The stock FBW departs.

Two bugs specific to the port showed how much the generic code had assumed:

- The asymmetric-load model read roll TVC as a load asymmetry and dropped the AoA cap to 12°. It now uses the symmetric nozzle moment.
- The engine-out yaw clamp counted only rudders, and the Ifrit has none. The yaw target froze, the splits sat idle and the jet departed with 64.6° of sideslip. Counting the tails and splits gives 1.1°.

Version 0.2.0 added camber coupling for the flaps and ailerons from each surface's chord fraction (K = 0.46 and 0.57), which raised roll at 135 kt from 106 to 151 °/s. It also added an MPO mode with separate roll and yaw axes and a buffed profile. Version 0.2.1 made the deceleron airbrake throttle-independent. An energy comparison on the same bench explains why the Ifrit "feels heavier": from 243 kt at full aft stick it pulls about 7 g and 24 °/s against the F-22E's 5 g and 17 °/s. At 65°, though, its stock post-stall drag costs about 38 kt per half second against 15 kt for the F-22E.

11 Discussion

Measured state over schedules. The laws that survived are the ones that read authority from the live airframe:

- the roll ceiling from online damping;
- the yaw-anchored roll;

- the low-thrust and asymmetric caps from authority sweeps;
- capture braking at live thrust;
- the airbrake fade from its live drag share.

They carried over to stock thrust, to damage and to a different aircraft (the Ifrit) without retuning. The stock-performance build matched the buffed one within 1–3%. Schedules that encoded authority indirectly, such as the 35–50° rudder fade and the flat low-IAS yaw margin, turned out wrong once measured.

Airframe ceiling versus FCS fault. A recurring question was whether a behaviour is the controller or the physics. The method that settled it was to check whether the surfaces deliver the FCS's own command:

- **Airframe ceiling:** a slow finless roll at the 40° cap follows its command, and the command is bounded by the 6–7 °/s of yaw rate the decelerons can make. The −60° corkscrew departures and VMC with no fins below 216 kt fall in the same class.
- **FCS faults:** the pitch bounce from the sideslip governor (2.2.20), the hidden VMC saturation (2.2.22) and the stab freeze (2.2.21).

Telling these apart honestly, including retracting a misattribution, mattered more than any single gain.

Open issues.

1. A bistable pitch dip of −14 to −22 °/s on some releases from rolling pulls, where one stab goes to its nose-down stop.
2. Finless held rolls below 35° AoA still depart. A roll rate limited by sideslip would be the fix.
3. With no fins the sideslip guard stops rolls for 1–3 s, because the decelerons take seconds to clear sideslip.
4. The source of the in-game stab roll mismatch is unknown.
5. Little of the post-2.2.15 work has been flown yet.

12 Conclusion

A plate-based game aero model can support a credible modern FCS if the controller works in the model's own terms. That means INDI, so the unmodelled airframe cancels out; an allocator that evaluates each surface on its real curve at its local AoA; and limits derived from measured authority. Most of the remaining difficulty is in the allocator's cost design and in recognising where the airframe, not the software, sets the limit. The same core now flies two very different aircraft.

A Release timeline (selected)

Ver.	Main change
1.4.0	Roll ceiling from online damping; monotone stab pitch model; sustained-rate unload; no nose-down with stick pulled
1.5.0	Static-margin rebalance (0.51 → 0.09 m); LERX 1.6, CG +0.05 m
1.6.0	Low-IAS stab roll cost; 20 ms actuator lag; 60 ms stick smoothing; outer flaperon droop
1.7.0	Low-IAS yaw-margin curve; yaw FF blend to 100% by 28°; override latch; AoA unwrap
1.7.6	Reversed-flow stab model; rudders back in reversed flow
1.8.0	Low-$\bar{q}$ allocator relief; inertia-aware roll limit; ground aero-brake; slats to 70°
1.8.14	Bench on the real airframe; coupling FF off; released-stick pitch priority; stab rate bound
1.8.17–20	AsymRollHeadroom; paired stab/TVC; LateralStabReserve; thrust-aware capture; anti-windup
1.8.34–35	Rudder sideslip analysis; sideslip-only rudder range
1.8.41	Pedal-in roll-fight cost
1.9.0	Hybrid INDI; command shaping; AoA-to-rate hand-over; low-thrust AoA cap
2.0.0	Staggered stall 25/35/45; flaperon 1.2, LERX 1.3
2.0.12	Damage lateral limits; yaw-target clamp; VMC limiter; fuel trim; takeoff stab split fix
2.1.0	G + pitch-rate law default; yaw-anchored roll; sideslip-rate lead
2.2.0	Rudder fade 50–60°; fin camber coupling
2.2.4	Negative AoA limit −50°
2.2.7	Rolling-pull sideslip governor; split-flaperon decelerons
2.2.14	TvcNeutralHold
2.2.15–18	StabParkEscape; released-stick fight cost; airbrake drag-share fade; stall-window search fix
2.2.19	FinLossAoACap 40; full deceleron yaw
2.2.20–22	Governor unload fix; stab split-only fight cost; VMC behind clamp
2.2.23	Governor defaults: sideslip margin 9°, roll trade 0.7
Ifrit 0.1.0	Port of 2.2.5: split and differential-TVC effectors
Ifrit 0.2.x	Rudderless yaw clamp; camber; MPO separate axes; buffed profile; throttle-independent airbrake

Data note. The airframe data extracted from Aryx's asset bundle are not reproduced here. All figures are derived results (bench outputs and telemetry regressions).